

Video Transcript

Video 13: Using Debezium with Spring Boot (06:39)

Next up, we're going to be creating the app that is going to be doing the change data capture. This is a very simple app, a Spring Boot app, your Hello World app, that is simply going to have one more block of code to be able to listen to changes in the database using the Debezium. As before, we're going to be going through three stages. The very first one is to create a Docker file. Then from it, we're going to create a Docker image and then finally we're going to create a container based on that image. The Docker file itself is going to be referencing that code, that Spring Boot code that I mentioned, and you can download that code from the following repository within GitHub.

You can see here the Docker file, a very simple basic Docker file that references Maiven, which is needed to be able to develop and run Spring Boot applications. Aside from that, you can see there that it's going to copy that application into the container into a directory called 'tmp'. Now here at the bottom, we have the command to be able to create the image and note the name that I'm giving the image. The image is going to be called the debeziumabel. So, let's go ahead and create that image.

But before I do so, let me go ahead and move to the third folder. That third folder has the files that are needed to create this application, but I'm going to start off here by creating by creating the image. And so as you can see there, that was pretty fast and that has now been created. Now once we have our image, then we're going to create our container. And let's take a look here at some of the arguments. You can see there that I'm going to call that container, I'm going to call it the debeziumserver. I'm then going to add this container to the same network that we have been using, network netabel, which stands for network abel.

And then I'm going to be basing this container on the debeziumabel image that we just created. And we're going to be opening that container to the bash. So, let's go ahead and clear our terminal here, and I'm going to go ahead and paste and run that command. And in a moment, we should open up the prompt. If I hit 'return', if I list my files, you can see there that I have help, I have pom, I have the source directory. And if I move within my host here into that third folder

and list my files, you can see there that I have the same.

Well, actually let me move into app because that has the same. And so now that we have created our container and we have moved our code into it, the next step is going to be to run the application. But right before we do so, let me go ahead, and show you that additional block of the Debezium code that was added. And so I'm going to show you that folder and I will do it here locally since I have smaller font and it won't be so overwhelming. And I'm going to move through that hierarchy of namespaces that Java applications have all the way into the 'Config' folder.

Then I'm going to list my files and that is the file I want. I'm going to go ahead and open it up with nano to get some syntax highlighting. And as you can see there, this is a pretty simple file. It simply references, imports some classes as you can see there. And then once we get to the code, that first block that has those dashes is simply figuring out the directories, and this second part is really instantiating that the Debezium configuration. And you can see here if you read through it, and I will leave that up to you, that we are connecting to MySQL Server, that we're specifying the port, that we're setting the password, that we're specifying the database that we're going to be watching and some more information.

But that's really all that this application is doing. So, it's simply going to be watching for changes on that database and its minimal code on top of a default configuration for a web application created through Spring Boot. So, I'm going to go ahead and exit from here and clear my console. And then I'm going to go ahead and run the application. And the command to run the application is the following. So, I'm going to go ahead and hit 'enter' here. And so as you can see there, the application is up and running.

Note that it has noticed that last insert or the only insert that we made into the database, you can see here John Doe, the email, and so on. And so you can see there that this is now up and running. So, let's go ahead and open up Workbench and we're going to go ahead and insert one more row. So, let's open up a new connection and let's go ahead and add that new entry. As you can see there, I'm inserting into 'customerdb' the database, the table, customer, the values of the ID of two, 'Peter Parker', and then 'peter@mit.edu'. So, let me go ahead and run that command. And as you can see there that was inserted. We get the verification here. And if we look at our server, you can see here that all of that information was captured. You can see here Peter Parker, you can see the email plus some additional information that is captured as part of that event. So, as you can see there, we have successfully created this application. And you

should know that Debezium is a very robust CDC, change data capture, application library that can be used and that has tremendous scalability. It has quite a bit of range. It has connectors for most of the widely used data stores. So, it's a great one to understand depth and to be able to learn.

